21-242 Recitation 12

Matrices — Q4. Composition corresponds to matrix multiplication

Course-level solution, with a later-course alternative when useful

October 1, 2026

Prompt

Prove that $[S \circ T]_{\mathcal{CA}} = [S]_{\mathcal{CB}}[T]_{\mathcal{BA}}$.

Course-level solution

Let $T : U \to V$ and $S : V \to W$, with bases $\mathcal{A}$ of U, $\mathcal{B}$ of V, $\mathcal{C}$ of W. For any $\vec{u} \in U$, Q3 three times gives

$$\begin{aligned}[S \circ T]_{\mathcal{CA}}\,[\vec{u}]_{\mathcal{A}} &= [(S \circ T)(\vec{u})]_{\mathcal{C}} = [S(T(\vec{u}))]_{\mathcal{C}} = [S]_{\mathcal{CB}}[T(\vec{u})]_{\mathcal{B}} \\ &= [S]_{\mathcal{CB}}[T]_{\mathcal{BA}}\,[\vec{u}]_{\mathcal{A}}.\end{aligned}$$

The coordinate map $U \to \mathbb{F}^{\dim U}$ is surjective, so two matrices that act the same on every coordinate vector are equal: $[S \circ T]_{\mathcal{CA}} = [S]_{\mathcal{CB}}[T]_{\mathcal{BA}}$.
(Directly on columns: the jth column of the left side is $[(S \circ T)(a_j)]_{\mathcal{C}} = [S]_{\mathcal{CB}}[T(a_j)]_{\mathcal{B}}$, which is the jth column of the product.)